

Crowd Density Alert Response

OPERATIONAL — UNRESTRICTED

TEXAS SMART CITY PUBLIC SAFETY PROGRAM

STANDARD OPERATING PROCEDURE

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AUTHORITY: TX-SCPS-POL-001 (AI Surveillance Deployment Policy), TX-SCPS-POL-002 (Crowd Management Data Governance), Texas Government Code §418 (Emergency Management)

Section 1 — PURPOSE

This Standard Operating Procedure (SOP) defines the mandatory step-by-step actions that operations centre staff and field units must execute when the AI-powered crowd density monitoring system generates a CRITICAL or EMERGENCY alert. It converts AI-generated sensor output into a structured human response workflow with defined responsibilities, time limits, and escalation paths.

Section 2 — SCOPE

This SOP applies to:

- City Operations Centre (OC) supervisors and watch managers
- Dispatch coordinators monitoring the Smart City Dashboard
- Field officers assigned to monitored zones (Austin, Dallas, Houston deployment areas)
- Zone commanders and sector supervisors
- Emergency Operations Centre (EOC) duty officers

Section 3 — DEFINITIONS

Term	Definition
CRITICAL Alert	AI system has detected person_count >= zone CRITICAL threshold for the monitored camera zone and the density_score >= 0.75 (severe crowding risk).
EMERGENCY Alert	person_count >= EMERGENCY threshold OR CRITICAL condition persists for >= 10 minutes without de-escalation.
Zone Response Protocol (ZRP)	Pre-planned resource deployment configuration for each monitored zone, maintained by the zone commander.
Density Score	Normalised value 0.0–1.0 produced by the DM-Count model representing crowd packing density relative to zone capacity. A score of 1.0 represents maximum observed density.
Safe Threshold	Zone-specific person_count below which no active response is required (see TX-SCPS-SOP-002 for per-zone values).
Incident Log	Mandatory contemporaneous record maintained in the case management system for all CRITICAL and EMERGENCY events.

Section 4 — TRIGGERING CONDITIONS

This SOP is activated when the Smart City Dashboard (accessible at <http://dashboard:8080> internally, or the externally mapped address) generates a CRITICAL alert for any monitored zone. Alert conditions are evaluated every 5 seconds by the AI pipeline background task.

Primary trigger:

- Alert severity field = "CRITICAL" in /api/v1/alerts response
- OR dashboard WebSocket (/ws/alerts) pushes severity "CRITICAL" event

Secondary trigger (manual activation):

- Operations supervisor may activate this SOP manually if field reports indicate crowding despite AI system unavailability (pipeline offline, stream dropped).

Section 5 — RESPONSE PROCEDURE

STEP 1 — IMMEDIATE ACKNOWLEDGEMENT (0–2 minutes from alert)

Responsible: OC Supervisor (on-duty watch manager)

1.1 Acknowledge the alert in the Smart City Dashboard: Navigate to Alerts panel, click "Acknowledge" on the CRITICAL alert. Dashboard calls PATCH /api/v1/alerts/{alert_id}/ack — confirm the alert status changes to acknowledged. Failure to acknowledge within 2 minutes triggers an auto-escalation notification to the sector supervisor.

1.2 Open the live camera feed for the flagged zone: Click the camera icon on the Map Panel to open the WHEP video stream. Visually confirm AI-reported crowding is accurate (rule out false positive: obstructed lens, lighting artefact, stream corruption).

1.3 If confirmed TRUE POSITIVE — proceed to Step 2. If suspected FALSE POSITIVE: Log the suspected false positive in the Incident Log with supporting rationale. Do NOT stand down without supervisor sign-off. Notify AI Operations on-call (see TX-SCPS-SOP-003 Contact Directory) to investigate inference accuracy. Continue to monitor at 1-minute intervals. If density persists for 5+ minutes with no physical cause, treat as TRUE POSITIVE.

1.4 Activate the Zone Response Protocol (ZRP) for the flagged camera zone: ZRPs are stored in the Ops Centre shared drive under /ZRP/<zone_id>/. Each ZRP specifies the pre-designated radio channel, field unit callsigns, and rendezvous points.

1.5 Notify the nearest available field unit via designated radio channel: Transmission format: "All units — Zone Alert [zone_name], camera [camera_id], density CRITICAL, person count [N]. Unit [callsign], respond and confirm ETZ (estimated time to zone)." Log transmission time in Incident Log.

STEP 2 — ESCALATION (2–5 minutes from alert)

Responsible: OC Supervisor + Dispatch Coordinator

2.1 If density persists at CRITICAL level at the 2-minute mark: Dispatch minimum 2 officers to the zone perimeter. Officers must establish a visible presence at zone access points before attempting crowd dispersal. Dispatch log entry must include: unit IDs, dispatch time, zone location coordinates, expected arrival.

2.2 Notify City Emergency Operations Centre (EOC) if either condition is met: a) person_count exceeds EMERGENCY threshold for the zone (see TX-SCPS-SOP-002). b) CRITICAL condition has persisted for >= 5 minutes without reduction. EOC notification channel: [configured in ZRP per jurisdiction]. Notification must include: zone ID, person count, duration of CRITICAL state, units dispatched, video stream reference.

2.3 If the event involves a pre-planned public gathering (festival, rally, parade): Cross-reference with event permit database before escalating. Notify event safety officer and request crowd management support from event stewards. Elevated crowd density may be expected; apply event-specific thresholds if documented in the ZRP.

2.4 OC Supervisor must brief sector commander within 3 minutes of EOC notification.

STEP 3 — ACTIVE MONITORING (ongoing, duration of incident)

Responsible: OC Supervisor + designated monitoring officer

3.1 Log all AI-reported density readings at 1-minute intervals during the incident: Source: GET /api/v1/analytics/density/{stream_id} — record person_count, density_score, severity per zone. Alternatively, pull from the heatmap panel or trend chart in the Planning Panel. All readings must be timestamped and entered into the Incident Log.

3.2 Re-evaluate severity every 5 minutes: If person_count remains CRITICAL: maintain current response posture. If person_count has decreased to HIGH: reduce to monitoring-only with one unit on standby. If person_count drops to SAFE threshold: proceed to de-escalation (Step 4).

3.3 Maintain radio contact with field units every 5 minutes. Record all unit status updates in the Incident Log.

3.4 Capture heatmap evidence: GET /api/v1/analytics/heatmap/{stream_id} — save base64 heatmap image to the Incident Log at 5-minute intervals for duration of the CRITICAL state.

3.5 If a secondary camera zone in the same location also escalates to CRITICAL simultaneously, treat as a multi-zone event and notify EOC immediately regardless of duration.

STEP 4 — DE-ESCALATION (when person_count < safe threshold)

Responsible: OC Supervisor

4.1 Confirm AI-reported density has returned below the zone safe threshold for at least two consecutive 5-second readings (10 seconds of sustained safe density).

4.2 Notify field units to stand down active response posture. Units may remain on the zone perimeter for 15 minutes to ensure no rebound.

4.3 Notify EOC of de-escalation and estimated stand-down time for dispatched resources.

4.4 Close the active alert in the dashboard. Log de-escalation timestamp.

STEP 5 — POST-INCIDENT DOCUMENTATION (within 60 minutes of de-escalation)

Responsible: OC Supervisor (primary) + responding officers (supporting)

5.1 Complete the Incident Log in the case management system:

- Incident ID (auto-generated or manually assigned)
- Zone ID and camera stream ID
- Alert start time, peak person_count, peak density_score
- Time of acknowledgement, dispatch, EOC notification, de-escalation
- Unit IDs of all responding officers
- Any false-positive determinations and supporting rationale

5.2 Preserve video evidence per TX-SCPS-POL-002 retention schedule:

- CRITICAL incidents: retain video for minimum 90 days.
- EMERGENCY incidents: retain indefinitely pending legal review clearance.
- File reference: record the MediaMTX stream URL and timestamp range in the Incident Log.

5.3 Submit incident summary to zone commander within 2 hours of de-escalation: Summary must include: timeline, peak metrics, response actions, personnel involved, any deviations from this SOP, and recommended follow-up.

5.4 Submit After-Action Report using TX-SCPS-TMPL-001 within 24 hours for:

- Any EMERGENCY-level event.
- Any CRITICAL event where EOC was notified.
- Any event involving crowd injuries, property damage, or officer use of force.

5.5 Flag any recurring CRITICAL alerts in the same zone within a 7-day period to the AI Operations team for threshold review (see TX-SCPS-SOP-002 for threshold adjustment procedure).

Section 6 — ESCALATION MATRIX

Severity	Duration	Action Required
CRITICAL	0–2 min	OC acknowledge, ZRP activate, field unit notify
CRITICAL	2–5 min	Minimum 2 officers dispatched
CRITICAL	5+ min	EOC notification mandatory
CRITICAL	10+ min	EMERGENCY declared, full mobilisation per ZRP
EMERGENCY	Immediate	EOC notification, sector commander briefed, all ZRP resources activated

Section 7 — COMMUNICATIONS PROTOCOL

- All radio transmissions related to a CRITICAL incident must reference the Incident ID.
- Do not use personal names or officer badge numbers over unencrypted radio channels.
- All EOC notifications must be via the secure EOC hotline or encrypted messaging channel.
- Media enquiries must be directed to the Public Information Officer (PIO); OC staff must not comment to media.

Section 8 — SYSTEM FAILURE CONTINGENCY

If the Smart City AI Dashboard becomes unavailable during an active CRITICAL incident:

- Revert to manual CCTV monitoring via backup display terminals.
- Continue all response steps based on visual assessment.
- Notify AI Operations on-call per TX-SCPS-SOP-003 immediately.
- Increase field unit patrol frequency to compensate for AI monitoring gap.
- Document the system outage period in the Incident Log.

Section 9 — TRAINING AND CERTIFICATION

All OC supervisors must complete annual training on this SOP, including:

- Tabletop exercise using simulated CRITICAL alert scenarios.
- Dashboard proficiency assessment.
- Radio protocol refresher.

Training records must be maintained in the personnel management system.

Section 10 — DOCUMENT CONTROL

Owner: Smart City Operations Division

Approver: Chief of Public Safety Operations

Next Review: 2027-04-01

Related Documents:

- TX-SCPS-POL-001 (AI Surveillance Deployment Policy)
- TX-SCPS-POL-002 (Crowd Management Data Governance)
- TX-SCPS-POL-003 (AI Operations Performance Standards)
- TX-SCPS-POL-004 (Privacy and Civil Liberties Safeguards)

- TX-SCPS-SOP-002 (Zone Severity Thresholds)
- TX-SCPS-SOP-003 (AI Pipeline Operational Runbook)
- TX-SCPS-TMPL-001 (Incident After-Action Report Template)

END OF DOCUMENT — TX-SCPS-SOP-001 v1.0